

Quick Check

Lesson 3: Where Do the Wastes Go? Shedding Old Leaves and Locking Away Crystals

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.3: Excretion in Plants | Lesson 3

Plants have no kidneys. A student predicts where a plant keeps wastes that cannot dissolve in water, such as calcium oxalate crystals, tannins and resins. Which prediction fits what the lesson develops?

- A** They dissolve in water and leave the plant as sweat through the leaves
- B** They are stored in the growing tips so the plant can reuse them quickly
- C** Plants do not make any wastes, so there is nothing to store
- D** They are stored in non-vital parts such as bark, old leaves, fruits and flowers, which are later shed

A group examines slides of a young green mango leaf and an old yellow mango leaf under a light microscope. They count needle-shaped crystals in 3 fields of view for each leaf. Which result would support the idea that old leaves store wastes?

- A** The same number of crystals in both leaves, as all leaves hold equal wastes
- B** Crystals only in the veins of the young leaf
- C** More crystals in the young green leaf than in the old yellow leaf
- D** More crystals in the old yellow leaf than in the young green leaf

In the leaf microscopy practical, learners add a drop of iodine to a second slide of leaf tissue. Why do they do this?

- A** To make the leaf tissue produce more crystals
- B** To dissolve calcium oxalate crystals so they can be counted
- C** To check for starch, which stains blue-black, so it is not mistaken for crystals
- D** To kill the cells so the crystals become bigger

A learner sees needle-shaped crystals in old leaf tissue. After adding a drop of ethanoic acid at the edge of the coverslip, the crystals do not dissolve. Which conclusion is best supported?

- A** The crystals are probably sugar, because sugar never dissolves
- B** The crystals are probably calcium oxalate, which does not dissolve in ethanoic acid
- C** The crystals are probably starch, because they stain blue-black
- D** The crystals are probably calcium carbonate, because they fizz in acid

One group looks carefully at slides of an old leaf and finds no crystals in any field of view. What should the group do?

- A** Draw crystals anyway so their result matches the other groups
- B** Throw away the slide and copy another group's results
- C** Report honestly that no crystals were seen, record it as data, and look in other places such as the veins or the leaf edge
- D** Decide that the old leaf stores no wastes at all

Groups find more calcium oxalate crystals in old yellow leaves than in young leaves. Which reasoning best explains why leaf fall counts as excretion?

- A** Wastes are locked in a harmless solid form in tissue the plant no longer needs, and they are removed when the leaf falls
- B** Leaf fall removes crystals from the young leaves, which are the main storage site
- C** The crystals dissolve in the yellow leaf and leave as droplets at the leaf edges
- D** Leaf fall only helps the plant to lose water vapour through stomata

Learners revise the class model of plant excretion. Which pair correctly matches a waste with how it is removed?

- A** Calcium oxalate crystals stored in old leaves: removed when the leaves are shed
- B** Droplets at leaf edges (guttation): deposited as crystals in bark
- C** Water vapour: removed when old leaves fall
- D** Resin: breathed out through the stomata as a gas

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).